



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to M.K.S.SERVICES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

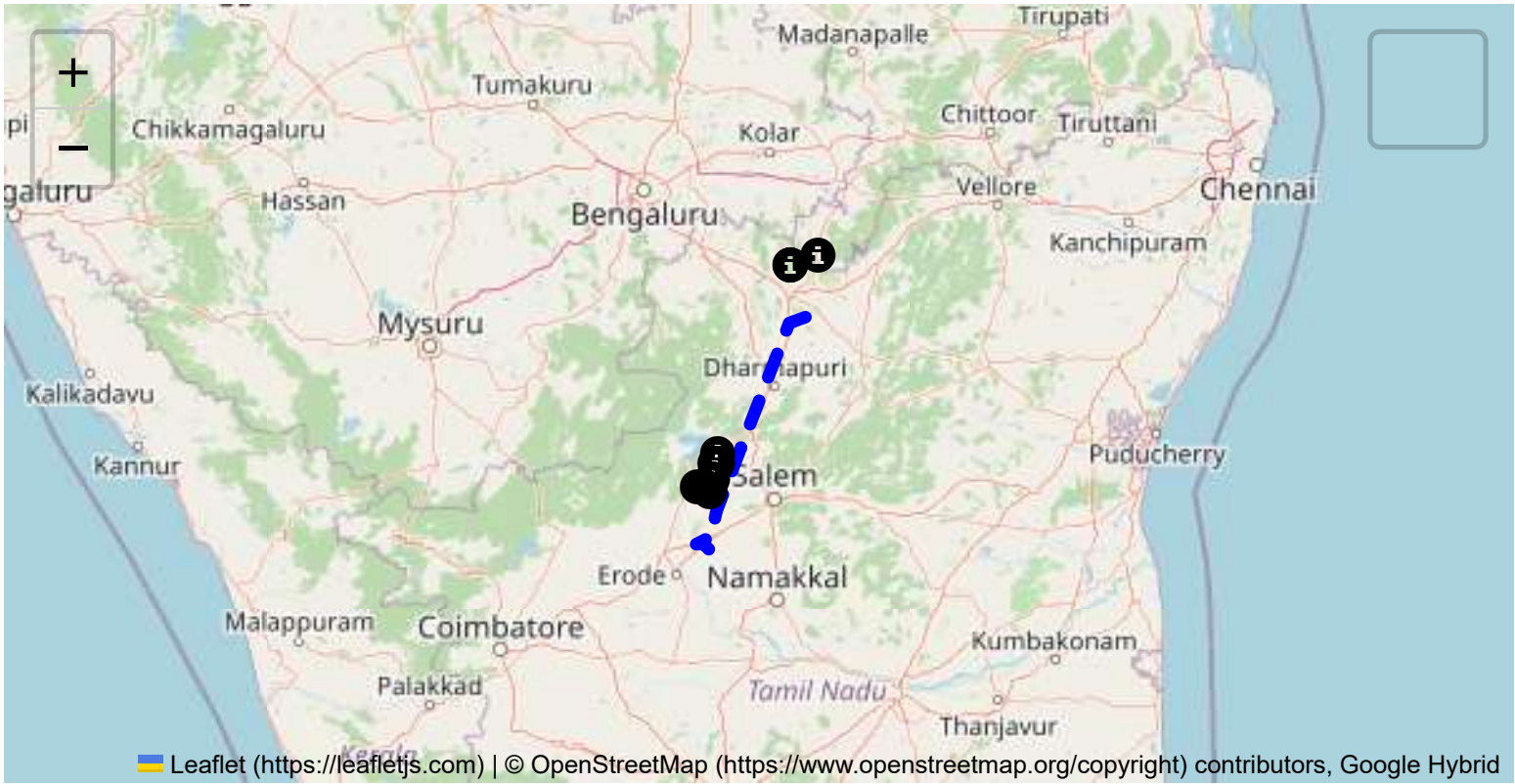
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 159.23 km**  
**Estimated Duration: 3.3 hours**  
**Adjusted Duration (Heavy Vehicle): 4.1 hours**  
**Start: (11.4381, 77.8734)**  
**End: (12.452232, 78.34832)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 2/400, Gandhinagar, Thogarapalli covers approximately 159.23 kilometers and includes various stops and junctions across Tamil Nadu state. This path primarily utilizes a mix of State Highways (SH) and Main Roads, traversing through small towns and villages, including major points like the Bhavani Main Road and SH 86.



## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The area can see heavy monsoon rains typically from October to December which may result in waterlogging and reduced visibility. Summer months (March to June) could bring high temperatures, contributing to potential road surface damage and vehicle overheating. Sporadic weather conditions like sudden cloudbursts and storms should be anticipated during the monsoon season.

## 3. Traffic Patterns

- **Peak Hours:** Traffic congestion is likely to happen during morning (7:30 AM - 9:30 AM) and evening (5:30 PM - 8:00 PM) hours, particularly around urbanized zones like Sankari, Erumaipatti, and Kaveripattinam.
- **Congestion-Prone Areas:** Bus stops and busy intersections around towns such as Sankari and Kaveripattinam can experience significant delays due to local traffic. Merchant activities and periodic local events in smaller villages might also temporarily increase congestion.

## 4. Assessment of Road Quality and Infrastructure

The infrastructure along this route includes varying quality roads. Town areas are generally well-paved, whereas rural stretches might have potholes and uneven road surfaces. Be cautious of narrow roads and weak bridge structures primarily in village areas. Limited road lighting may pose additional hazards at night.

## 5. Suggestions for Alternative Routes for Emergencies

In emergencies, an alternative route via Salem, using the NH44, could offer more direct access to medical facilities and police stations but may increase travel distance and duration. Diversions to more urbanized corridors might provide better infrastructure support.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to safety regulations as per local enforcement. This includes ensuring properly equipped vehicles, secured loading, and necessary documentation. Areas near schools and hospitals may have restricted transit hours or speed limits.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historically, there have been incidents involving vehicle breakdowns and accidents on the SH 86 due to mechanical failure or driver error. Most reports do not indicate severe hazardous material incidents, but precautionary measures are suggested due to frequent vehicle density on state highways.

## 8. Environmental Considerations and Sensitive Areas

The route crosses regions near reserved lands and agricultural areas which might have restrictions on material transport due to environmental policies. Noise and emission regulations are present in small towns to limit disturbance to residential zones.

## 9. Analysis of Communication Coverage

Telecom networks are generally robust along highways and town centers, but certain rural sectors between villages like Devannagoundanur might experience weak signals. Keeping communication devices charged and alternative contact methods is advisable.

## 10. Estimated Emergency Response Times

- Urban Areas:** Response times could be within 15-30 minutes.
- Rural Areas:** Response times might stretch to 45 minutes to an hour due to less accessibility. Having emergency contact numbers for nearby towns, along with clear transportation plans, enhances readiness.

## 11. Overall Summary of Risk Assessment

The route from Sangagiri to Gandhinagar involves medium risk for transporting hazardous materials, primarily due to variable road conditions and weather impacts. With careful planning around peak traffic times, weather checks, and alternative routing, risks associated with this journey can be reasonably managed. Prioritizing maintenance checks of transport vehicles and driver readiness are imperative for mitigating potential hazards. Regular training on new regulations and emergency procedures will further ensure smoother transit across this corridor.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 1 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 2 | Turn      | High       | 11.44900, 77.87406 | 15 KM/Hr    |
| 3 | Turn      | High       | 11.45352, 77.85698 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.45764, 77.85219 | 30 KM/Hr    |
| 5 | Turn      | High       | 11.46303, 77.84945 | 15 KM/Hr    |
| 6 | Turn      | High       | 11.45571, 77.81702 | 15 KM/Hr    |
| 7 | Turn      | Medium     | 11.45654, 77.81706 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 8  | Turn       | Medium     | 11.46916, 77.82431 | 30 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.49395, 77.88537 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.49465, 77.88488 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.56096, 77.89931 | 30 KM/Hr    |
| 12 | Turn       | Medium     | 11.57649, 77.90610 | 30 KM/Hr    |
| 13 | Turn       | Medium     | 11.59214, 77.90656 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.63833, 77.94818 | 30 KM/Hr    |
| 15 | Turn       | High       | 11.64243, 77.95198 | 15 KM/Hr    |
| 16 | Turn       | Medium     | 11.64721, 77.95118 | 30 KM/Hr    |
| 17 | Turn       | Medium     | 11.67870, 77.96846 | 30 KM/Hr    |
| 18 | Turn       | High       | 11.70227, 77.97864 | 15 KM/Hr    |
| 19 | Turn       | Medium     | 11.71416, 77.99604 | 30 KM/Hr    |
| 20 | Turn       | Medium     | 11.71715, 77.99969 | 30 KM/Hr    |
| 21 | Turn       | Medium     | 11.71921, 78.01133 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.72737, 78.02248 | 30 KM/Hr    |
| 23 | Turn       | High       | 11.73775, 78.03465 | 15 KM/Hr    |
| 24 | Turn       | High       | 11.73651, 78.03878 | 15 KM/Hr    |
| 25 | Turn       | High       | 11.74189, 78.04660 | 15 KM/Hr    |
| 26 | Turn       | Medium     | 11.78039, 78.05792 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.86753, 78.08331 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.86981, 78.08387 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.88041, 78.08143 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.91302, 78.05856 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.92526, 78.06488 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.93402, 78.06354 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.94600, 78.04740 | 30 KM/Hr    |
| 34 | Turn       | Medium     | 11.95625, 78.04748 | 30 KM/Hr    |
| 35 | Turn       | High       | 11.95880, 78.04989 | 15 KM/Hr    |
| 36 | Turn       | High       | 11.96040, 78.04947 | 15 KM/Hr    |
| 37 | Turn       | High       | 11.96320, 78.05139 | 15 KM/Hr    |
| 38 | Turn       | Medium     | 11.96463, 78.05077 | 30 KM/Hr    |
| 39 | Turn       | High       | 11.96597, 78.04920 | 15 KM/Hr    |
| 40 | Turn       | Medium     | 11.98504, 78.06287 | 30 KM/Hr    |
| 41 | Turn       | High       | 12.24611, 78.19519 | 15 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 42 | Turn       | Medium     | 12.25200, 78.20034 | 30 KM/Hr    |
| 43 | Blind Spot | Blind Spot | 12.40692, 78.21762 | 10 KM/Hr    |
| 44 | Turn       | Medium     | 12.40628, 78.22039 | 30 KM/Hr    |
| 45 | Turn       | Medium     | 12.40691, 78.22149 | 30 KM/Hr    |
| 46 | Turn       | High       | 12.40667, 78.22267 | 15 KM/Hr    |
| 47 | Blind Spot | Blind Spot | 12.41749, 78.22507 | 10 KM/Hr    |
| 48 | Turn       | High       | 12.39685, 78.30841 | 15 KM/Hr    |
| 49 | Turn       | Medium     | 12.39490, 78.31233 | 30 KM/Hr    |
| 50 | Turn       | High       | 12.39932, 78.33507 | 15 KM/Hr    |
| 51 | Turn       | High       | 12.41856, 78.33140 | 15 KM/Hr    |
| 52 | Turn       | Medium     | 12.42231, 78.33446 | 30 KM/Hr    |
| 53 | Turn       | High       | 12.42453, 78.33510 | 15 KM/Hr    |
| 54 | Turn       | High       | 12.43365, 78.34494 | 15 KM/Hr    |
| 55 | Turn       | High       | 12.43413, 78.34578 | 15 KM/Hr    |
| 56 | Turn       | Medium     | 12.44387, 78.34749 | 30 KM/Hr    |
| 57 | Turn       | Medium     | 12.44472, 78.34837 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                              | coordinates            | speed_limit | risk_level |
|----|----------|-----------------------------------|------------------------|-------------|------------|
| 1  | hospital | Sri Shanthi Hospital              | 11.569272, 77.901937   | 30 km/h     | Medium     |
| 2  | hospital | Sugumar Hospital                  | 11.7396031, 78.0426955 | 30 km/h     | Medium     |
| 3  | hospital | Omalur Government Hospital        | 11.74161, 78.044296    | 30 km/h     | Medium     |
| 4  | clinic   | R.R. Nursing Home                 | 11.7394129, 78.0439861 | 30 km/h     | Medium     |
| 5  | hospital | Sri Hari Hospital                 | 11.7396726, 78.0440701 | 30 km/h     | Medium     |
| 6  | hospital | Sri Sugam Hospital                | 11.7429821, 78.0461039 | 30 km/h     | Medium     |
| 7  | hospital | Saravana Hospital                 | 11.7409303, 78.0458308 | 30 km/h     | Medium     |
| 8  | hospital | Arun Hospital                     | 11.7429892, 78.0466881 | 30 km/h     | Medium     |
| 9  | clinic   | Gomathi Clinic                    | 11.7445293, 78.0470401 | 30 km/h     | Medium     |
| 10 | clinic   | Saratha Clinic                    | 11.7422818, 78.0458175 | 30 km/h     | Medium     |
| 12 | clinic   | Sri Meenakshi Nursing Home, Salem | 11.7449162, 78.0474637 | 30 km/h     | Medium     |
| 13 | hospital | Vinayaka Child Hospital           | 11.745812, 78.0480225  | 30 km/h     | Medium     |

|    | type     | name                               | coordinates            | speed_limit | risk_level |
|----|----------|------------------------------------|------------------------|-------------|------------|
| 14 | hospital | Dr. G.R.Thilakam Hospital          | 11.746097, 78.04769    | 30 km/h     | Medium     |
| 15 | clinic   | Divyam Child Care Clinic           | 11.7447135, 78.0473858 | 30 km/h     | Medium     |
| 16 | hospital | Dr. Nagaraj Hospital               | 11.7464035, 78.0472401 | 30 km/h     | Medium     |
| 17 | hospital | Thulasi Hospital                   | 11.7461516, 78.0469984 | 30 km/h     | Medium     |
| 18 | clinic   | Momota Clinic                      | 11.744982, 78.047397   | 30 km/h     | Medium     |
| 19 | hospital | Amirthalingam Hospital             | 11.7456708, 78.0470548 | 30 km/h     | Medium     |
| 20 | hospital | Children Hospital                  | 11.7458377, 78.0480205 | 30 km/h     | Medium     |
| 21 | clinic   | Shri Ramana Poly Clinic            | 11.7477122, 78.0479293 | 30 km/h     | Medium     |
| 23 | hospital | Anugraha Multi Speciality Hospital | 11.773549, 78.0546408  | 30 km/h     | Medium     |
| 25 | clinic   | JSA Clinic                         | 11.8700619, 78.0844982 | 30 km/h     | Medium     |
| 26 | hospital | L.R. Hospital                      | 11.8703354, 78.0844554 | 30 km/h     | Medium     |
| 27 | hospital | Thoppur,Government Hospital        | 11.9398824, 78.0551658 | 30 km/h     | Medium     |
| 29 | hospital | Government Hospital                | 12.3048002, 78.2060881 | 30 km/h     | Medium     |

### Crowded Spots

|    | type    | name                                                 | coordinates               | speed_limit | risk_level |
|----|---------|------------------------------------------------------|---------------------------|-------------|------------|
| 0  | school  | KRP Matric. Hr. Sec School                           | 11.4546193,<br>77.8142445 | 30 km/h     | Medium     |
| 11 | school  | Fathima Hr Sec School                                | 11.7410508,<br>78.0479272 | 30 km/h     | Medium     |
| 22 | school  | John Britto Matriculation Higher<br>Secondary School | 11.7676907,<br>78.0513092 | 30 km/h     | Medium     |
| 24 | college | Dhirajlal Gandhi College of Technology               | 11.7968014,<br>78.0562095 | 30 km/h     | Medium     |
| 28 | college | Laxminarayana Polytechnic College                    | 12.0421277,<br>78.103203  | 30 km/h     | Medium     |
| 30 | school  | Government Boys Higher secondary<br>school           | 12.3047059,<br>78.2065495 | 30 km/h     | Medium     |

### Route Photos of Risky Spots





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44029, 77.87544**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.44900, 77.87406**





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45352, 77.85698**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45764, 77.85219**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.46303, 77.84945**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45571, 77.81702



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45654, 77.81706





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46916, 77.82431



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.49395, 77.88537



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.49465, 77.88488





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.56096, 77.89931



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.57649, 77.90610





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.59214, 77.90656



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.63833, 77.94818



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.64243, 77.95198





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.64721, 77.95118



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.67870, 77.96846





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.70227, 77.97864**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.71416, 77.99604**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.71715, 77.99969**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.71921, 78.01133



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.72737, 78.02248





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.73775, 78.03465**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.73651, 78.03878**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.74189, 78.04660**





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.78039, 78.05792



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.86753, 78.08331





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.86981, 78.08387



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.88041, 78.08143



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.91302, 78.05856





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.92526, 78.06488



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.93402, 78.06354





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.94600, 78.04740



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.95625, 78.04748



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.95880, 78.04989





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96040, 78.04947



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96320, 78.05139





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.96463, 78.05077



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.96597, 78.04920



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.98504, 78.06287





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 12.24611, 78.19519



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.25200, 78.20034





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 12.40692, 78.21762



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.40628, 78.22039



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 12.40691, 78.22149





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.40667, 78.22267



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 12.41749, 78.22507





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 12.39932, 78.33507

